Python

Course Syllabus

Class: Python with Gen @ 7:30 AM (IST) by Mr. Satish Gupta-[24-06-2026] | Duration: 26 Jun 2026 - 31 Oct 2026

A beginner-friendly Python course covering fundamentals, logic building, and real-world examples.

COURSE TOPICS

1. Introduction to Python

- Introduction to Python
- Python Syntax
- Python Installation and IDE Usage

2. Statements, Keywords, Comments and Usage

- Statements
- Keywords
- Comments and Usage

3. Variables, Identifiers

- Variables and Memory Allocation
- Rules for Variable Naming
- Multiple assignment, Reassignment, Swapping, Deleting Variables

4. OOPS

- Constructor
- Introduction to OOPS
- Classes
- Objects
- Methods

- Attributes
- Access Specifiers
- Encapsulation
- Class and Instance Variables
- Inheritance
- Single Inheritance
- Usage of Super Method
- Multiple Inheritance
- Multi Level Inheritance
- Hierarchical Inheritance
- Hybrid Inheritance
- Polymorphism
- Method Overloading
- Method Overriding
- Operator Overloading
- Data Abstraction
- Super Method

5. Operators

- Arithmetic Operators
- Relational/Comparison Operators
- Logical Operators
- Assignment Operators
- Membership Operators
- Identity Operators
- Bitwise Operators

6. Data Types

- Introduction to Data Types
- Numeric (int, float, complex)
- Text (str)
- Sequence (list, tuple, range)
- Mapping (dict)
- Set (set, frozenset)
- Boolean (True, False)
- Binary (bytes, bytearray, memoryview)

7. Type Conversions

- Introduction to Type Convertions
- Types of Type Convertions

8. Input Statements

- Taking String Data
- Taking Integer Data
- Taking List Data
- Taking input from users
- Taking Tuple Data
- Taking Set Data
- Taking Dictionary Data

9. Output Statements

- Using Old Style Formatting [using Commas]
- Using Modulo Operator
- Using F- Format Method
- Using Dot Format Method

10. Conditional Statements

- IF
- IF - ELSE
- ELIF
- Nested Conditional Statements
- Match Case
- Problem Solving on Conditional Statements

11. Control Statements

- For Loop
- While Loop
- For With Else
- While With Else
- Problem Solving on Control Statements
- Nested Loops
- Problem Solving on Patterns

12. Strings

- Introduction to Strings
- Operations on Strings
- Functions within Strings

13. Lists

- Introduction to Lists
- Operations on Lists
- Functions within Lists
- Creation
- Accessing
- Insertion
- Modification
- Deletion
- List Comprehension
- Nested Lists
- Concatenation
- Repetition

14. Tuple

- Introduction to Tuples
- Functions within Tuples
- Operations on Tuples
- Creation
- Accessing
- Concatenation
- Repetition
- Nested Tuples

15. Dictionary

- Introduction to Dictionary
- Operations on Dictionary
- Functions within Dictionaries

16. Sets

- Introduction to Sets
- Operations on Sets
- Functions within Sets

17. Functions

- Definition of Functions
- User Defined Functions
- Function Arguments (Positional;Default;Keyword arguments)
- Function Arguments (Variable length;Keyword Variable length arguments)

- Scope of the Variables
- Pass By Value & Pass By Reference
- Built in Functions
- Recursive Functions
- Anonymous Functions
- Map Function
- Filter Function
- Reduce Function

18. Decorators

- Introduction to Decorators
- Custom Decorators
- Decorators with Arguments
- Built-in Decorators

19. Generators

- How Generators works
- Yield and Next Keywords in Generators
- Difference b/w Functions & Generators

20. Modules

- Intro to Modules
- Built-in Modules
- User Defined Module

21. Packages

- Creating and Importing Packages
- Understanding init.py
- Nested Packages and Code Organization

22. Exception Handling

- Introduction to Handling Errors
- Except
- Try
- Else
- Finally

23. Files & Directories

- File Operations
- Open, Read, write, append and close
- Directories
- Create Directory, Remove Directory , Represent files in Directory
- Append

24. Regular Expressions

- Introduction to Regular Expression
- Understanding Regular Expression
- Meta Characters usage

25. Date and Time

- Date Methods
- Time Methods

26. Garbage Collection

- Introduction to Garbage Collection
- Reference Counting and Cyclic References
- Using the gc Module

27. Python with Databases

- Introduction to Database Connectivity
- Working with Oracle/SQL
- Executing Queries using Python
- CRUD Operations
- Handling Exceptions in Database Operations
- Introduction to MySQL in Python
- CRUD Operations using MySQL
- MySQL Error Handling and Transactions

28. Multithreading & Multiprocessing

- Introduction to Multithreading
- Creating Threads
- Introduction to Multiprocessing
- Using Process, Pool, Queue
- Synchronization Techniques

29. Network Programming

- Introduction to Socket Module
- Creating Sockets
- Socket Communication
- Socket Management
- Client-Server Connection

30. GUI Programming with Tkinter

- Introduction to GUI and Tkinter
- Basic Widgets (Label, Button, Entry, Text, etc.)
- Layout Management (pack, grid, place)
- Event Handling and Commands
- Tkinter Projects (like Calculator, Form UI, etc.)

31. NumPy

- Introduction to NumPy & Installation
- Creating and Manipulating Arrays
- Array Indexing, Slicing & Iteration
- Array Operations & Broadcasting
- Random Module & Advanced Utilities

32. Pandas

- Introduction to Pandas
- Creating and Manipulating Series
- Creating and Manipulating DataFrames
- Data Selection, Filtering, and Indexing
- Basic Data Analysis and Summary Statistics

33. Matplotlib

- Introduction to Matplotlib
- Creating Different Types of Plots
- Saving and Exporting Plots
- Figure and Axes Customization